

Avoiding the Student→Teacher Inverse in Cross-Latent On-Policy Distillation

Working Analysis (Draft)

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Abstract

This is a working (temporary) companion note to `cross_latent_distillation_problem.tex`. In the cross-latent setting with $d_A > d_B$ (the teacher VAE has more channels than the student VAE), the forward transport $T : \mathcal{Z}_A \rightarrow \mathcal{Z}_B$ (teacher→student, high→low) is easy and well-learned, whereas the inverse $T^{-1} : \mathcal{Z}_B \rightarrow \mathcal{Z}_A$ (student→teacher, low→high) is underdetermined and hard. We ask whether on-policy distillation can be carried out *using only the well-learned forward map* T , bypassing T^{-1} . The answer is yes, and the reason is structural: the inverse is required *only* because we insist on a *pointwise vector-field* target evaluated at student-chosen states. Measures push forward canonically under any measurable map, and differentiation of losses built from forward maps uses their *adjoints* (vector–Jacobian products), which always exist. Reformulating the objective in either of these two ways removes T^{-1} entirely. We give the precise argument and three concrete routes, and map them onto the existing L0/L1 structure of XOPD.

1 Setup and Notation

We reuse the notation of the parent note. Two VAEs induce latent spaces $\mathcal{Z}_A \subseteq \mathbb{R}^{d_A}$, $\mathcal{Z}_B \subseteq \mathbb{R}^{d_B}$ with encoders/decoders $(\mathcal{E}_A, \mathcal{D}_A)$, $(\mathcal{E}_B, \mathcal{D}_B)$ over a shared pixel space $\mathcal{X}$. The teacher is frozen with velocity v_A on $\mathcal{Z}_A$; the student has trainable velocity v_θ on $\mathcal{Z}_B$. Empirically

$$d_A > d_B,$$

so the forward transport

$$T : \mathcal{Z}_A \rightarrow \mathcal{Z}_B \quad (\text{teacher} \rightarrow \text{student, high} \rightarrow \text{low})$$

is a submersion (many-to-one), easy and well-learned; its would-be inverse

$$T^{-1} : \mathcal{Z}_B \rightarrow \mathcal{Z}_A \quad (\text{student} \rightarrow \text{teacher, low} \rightarrow \text{high})$$

is a dimensional expansion, underdetermined and hard. On-policy distillation evaluates its loss on the student-visited distribution ρ_θ (the marginal of a student rollout $\{z_B(t)\}$ under v_θ).

Where the inverse enters. The pushforward velocity field is, formally,

$$(T\#v_A)(z_B, t) = \mathbf{J}_T(T^{-1}(z_B)) v_A(T^{-1}(z_B), t). \quad (1)$$

Using (1) as a *pointwise target* at a student state $z_B \sim \rho_\theta$ requires a preimage $T^{-1}(z_B)$: this, and only this, is what forces the hard inverse.

2 Why the Inverse Is Hard, and When It Is Truly Needed

Definition 1 (Fiber and section). For $z_B \in \mathcal{Z}_B$ the *fiber* is $\mathcal{F}(z_B) := \mathsf{T}^{-1}(z_B) = \{z \in \mathcal{Z}_A : \mathsf{T}(z) = z_B\}$, a $(d_A - d_B)$ -dimensional set when T is a full-rank submersion. A *section* of T is a right inverse $s : \mathcal{Z}_B \rightarrow \mathcal{Z}_A$ with $\mathsf{T} \circ s = \text{id}_{\mathcal{Z}_B}$; equivalently $s(z_B) \in \mathcal{F}(z_B)$. Learning T^{-1} is exactly learning a section.

Proposition 1 (Vector-field pushforward is generally ill-defined under a non-injective map). Let T be a submersion with $d_A > d_B$. The expression $z_B \mapsto \mathbf{J}_{\mathsf{T}}(z) v_A(z)$ is independent of the choice of representative $z \in \mathcal{F}(z_B)$ *if and only if* v_A is T -projectable, i.e. the value $\mathbf{J}_{\mathsf{T}}(z) v_A(z)$ is constant along each fiber. In general it is not; hence $\mathsf{T}\#v_A$ is *not* a well-defined vector field, and a pointwise target (1) is only defined *after* fixing a section $s = \mathsf{T}^{-1}$.

Remark 1 (The real difficulty is the formulation, not network capacity). Prop. 1 says the pointwise velocity target is ill-posed *as a mathematical object*: different sections give different targets, and the “correct” section must recover teacher-exclusive fiber coordinates that z_B does not determine (the underdetermination of the parent note). Enlarging the network for T^{-1} therefore has diminishing returns; the leverage is in changing the objective so that no section is required.

Proposition 2 (Measures always push forward). For *any* measurable $\mathsf{T} : \mathcal{Z}_A \rightarrow \mathcal{Z}_B$, the pushforward measure $\mathsf{T}\#p_A$, defined by $(\mathsf{T}\#p_A)(E) = p_A(\mathsf{T}^{-1}(E))$, exists and is unique. In particular the projected-teacher marginal $\mathsf{T}\#p_A^0$ (and its noised versions along the flow) is well-defined with no choice of section.

Observation 1 (The escape principle). Combining Prop. 1 and Prop. 2: replace the *pointwise vector-field* target by a *distributional* target (which pushes forward canonically under T), and/or measure the discrepancy in a space reachable by *forward* maps so that gradients use adjoints (VJPs). Both differentiation and expectation then use only T (and transposes of forward maps), never the set-inverse T^{-1} .

3 Route A: Distribution / Score Matching (DMD/VSD style)

Replace the pointwise L1 target by matching the student’s on-policy marginal to the projected teacher marginal. Let $q_t := (\mathsf{T}\#p_A)(\cdot, t)$ (well-defined by Prop. 2) and let $\rho_{\theta,t}$ be the student’s rollout marginal at time t . Minimize a reverse-KL / score objective

$$\min_{\theta} \mathbb{E}_t [w(t) \text{KL}(\rho_{\theta,t} \parallel q_t)], \quad \nabla_{\theta} \approx \mathbb{E}_t \mathbb{E}_{z_B \sim \rho_{\theta,t}} \left[(\mathbf{s}_{\theta}(z_B, t) - \mathbf{s}_q(z_B, t)) \partial_{\theta} z_B \right], \quad (2)$$

where $\mathbf{s}_{\theta} = \nabla \log \rho_{\theta,t}$ (the “fake” score) and $\mathbf{s}_q = \nabla \log q_t$ (the “real”/target score). Both scores are evaluated *at student states* $z_B \in \mathcal{Z}_B$: no inverse. The target score $\mathbf{s}_q$ is obtained by fitting a small student-space score model on samples $z_B^0 = \mathsf{T}(z_A^0)$, $z_A^0 \sim p_A^0$ (teacher samples pushed forward through the *easy* T).

- **On-policy:** yes ($z_B \sim \rho_{\theta}$).
- **Uses T^{-1} :** no. Only forward T (to build target samples).
- **Cost:** one auxiliary score network; adversarial-style two-timescale training.
- **Why principled:** sidesteps Prop. 1 entirely by targeting a measure rather than a vector field.

4 Route B: Shared Pixel Space via Adjoint (VJP, not inverse)

Both decoders map into the shared pixel space $\mathcal{X}$. Measure the on-policy discrepancy after a *forward* decode of the student’s clean estimate $\hat{z}_B^0(z_B, t)$, with a stop-gradient teacher target:

$$L(\theta) = \mathbb{E}_{z_B \sim \rho_\theta} \left\| \mathcal{D}_B(\hat{z}_B^0) - \text{sg}[x^*(z_B, t)] \right\|^2, \quad x^* := \mathcal{D}_A(\Psi_A^0(\mathcal{E}_A(\mathcal{D}_B(\hat{z}_B^0)), t)), \quad (3)$$

where Ψ_A^0 is the teacher’s clean-sample estimator (one teacher query). The teacher target uses the *frozen* composition $\mathcal{E}_A \circ \mathcal{D}_B$; this is *not* a learned low→high network—the encoder $\mathcal{E}_A$ legitimately produces d_A channels from the decoded image. Crucially, the θ -gradient flows back only through the forward map $\mathcal{D}_B$ via its *vector–Jacobian product* $\mathbf{J}_{\mathcal{D}_B}^\top$, which always exists; nowhere is T inverted.

- **On-policy:** yes.
- **Uses T^{-1} :** no. Only forward/frozen maps and their adjoints.
- **Cost:** a decode+encode through two VAEs per step; round-trip VAE loss δ .
- **Why principled:** an x_0 (clean-sample) target avoids the projectability issue of Prop. 1; differentiation uses $\mathbf{J}^\top$ (adjoint), not the set-inverse.

5 Route C: A T -Anchored Minimal-Norm Section

If a pointwise velocity target is desired, do *not* learn a free T^{-1} . Anchor it to the already-good T by the minimal-norm section (Moore–Penrose pseudo-inverse in the affine case $\mathsf{T} = Az + b$):

$$\mathsf{T}^{-1}(z_B) := A^+(z_B - b), \quad AA^+ = \text{id}_{\mathcal{Z}_B}, \quad (4)$$

subject only to the right-inverse constraint $\mathsf{T} \circ \mathsf{T}^{-1} = \text{id}$ (not a full cycle). The teacher-exclusive fiber coordinates are set to their conditional mean; a small learned residual, optionally conditioned on the student DiT hidden states h_θ (parent note, §“Conditional Pullback”), refines them: $\mathsf{T}_c^{-1}(z_B) = A^+(z_B - b) + g_\phi(z_B, h_\theta)$ with g_ϕ zero-initialized.

- **On-policy:** yes; keeps the existing pointwise formulation.
- **Uses T^{-1} :** yes, but reduced from “a free low→high network” to “ T -anchored pseudo-inverse + small residual”.
- **Cost:** minimal; reuses T ’s parameters.

6 Mapping onto XOPD’s L0/L1

- **L0 (clean-sample regression) is already inverse-free:** it only needs sample transport $z_B^0 = \mathsf{T}(z_A^0)$ through the easy forward T .
- **L1 (on-policy transition / mean matching) is where T^{-1} bites:** it needs the teacher’s opinion at student states. Replace it by Route A (student-space score difference) or Route B (pixel x_0 matching) to make L1 depend only on the forward T ; or use Route C to keep L1 pointwise but with a cheap anchored inverse.

Route	On-policy	Uses T^{-1}	Key cost	Principle
A: score / DMD	yes	no	aux. score net, 2-timescale	measure pushforward
B: pixel adjoint	yes	no	$2\times$ VAE per step, δ	VJP of forward maps
C: anchored pseudo-inv.	yes	yes (cheap)	minimal	T-anchored section

Table 1: Three ways to reduce or remove the hard student $\rightarrow$ teacher inverse in on-policy L1.

7 Recommendation

- **Minimal change, runs now:** Route C (pseudo-inverse + hidden-state residual), keeping the current pointwise formulation.
- **Cleanest, truly eliminates the inverse:** Route A (score/DMD), which converts the ill-posed vector-field-inversion into a well-posed measure-matching problem while staying on-policy.
- **Middle ground:** Route B, on-policy and inverse-free, at the price of per-step VAE round-trips.

One-line takeaway. The inverse T^{-1} is required only because we chose a *pointwise vector-field* target at student-chosen states. Switching to distributional / score / x_0 objectives—which push forward canonically under the easy T and whose gradients use only adjoints (VJPs) of forward maps—removes T^{-1} from the problem.

8 Related: Cross-Space Bridge as a Route A interface

Nguyen et al. (arXiv:2606.32020) train a lightweight Bridge $\mathcal{B}_\phi : \mathcal{Z}_B \rightarrow \mathcal{Z}_A$ (frozen Student-decoder spatial prior + SwinIR projector; ℓ_1 latent recon + reverse-KL on Teacher attention) so that compact Students can run DMD2-style VSD/GAN *in Teacher space* under Cross-Resolution and Cross-VAE mismatch.

In the language of this note, Bridge is a concrete learned T^{-1} used as a **Route A interface**: Student samples are mapped into $\mathcal{Z}_A$, then score / adversarial matching proceeds where the Teacher is defined. It does not construct a forward T with Jacobian for pointwise velocity pushforward in $\mathcal{Z}_B$ (Routes B/C and XOPD L1 remain separate). Companion write-up: `docs/xopd/cross_space_distillation_bridge.tex`; inverse-method slot: `inverse_mapping_methods.tex` §Method 3.